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Lightning talk 10 | School of Science and Technology

Hackathons as a Medium for Disseminating Agentic AI in Research

Daniel Sikar and Dave Muir

How hackathons can disseminate agentic AI in research, education and workforce upskilling, illustrated through the Clinical AI Hackathon.

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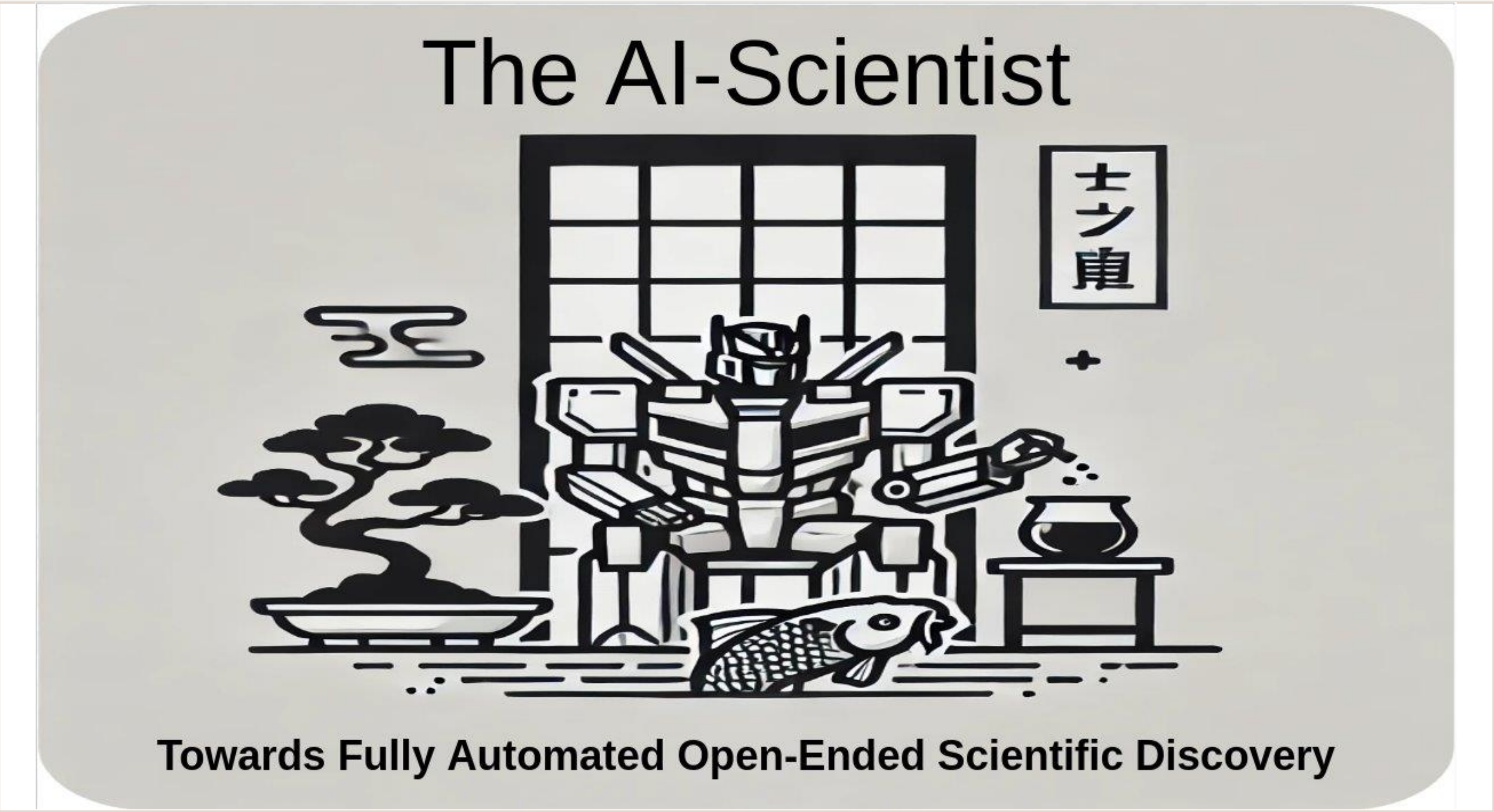
Agentic AI, where did it all begin?



Agentic AI, where did it all begin?

Paper

ArXiv - [Submitted on 12 Aug **2024** (v1), last revised 1 Sep 2024]
https://arxiv.org/abs/2408.06292

Company website
SakanaAI - https://sakana.ai/

News
"Japan's NVIDIA-backed **Sakana AI raises $214m**"
https://www.techinasia.com/news/japans-nvidiabacked-sakana-ai-raises-214m

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What is the AI Scientist?

Purpose: The AI Scientist is designed to be the first comprehensive system for **fully automated scientific discovery**

Functionality: It can generate **novel** research **ideas** (**via Semantic Scholar API queries**), **write code** (**Aider**, same backbone as **Coder**), execute experiments (via Python), visualize results, and author full scientific papers, complete with peer review, all autonomously.

Templates: **Provides three** foundational **templates** for research in different domains: NanoGPT for transformer-based tasks, 2D Diffusion for generative model performance, and Grokking for studying generalization in neural networks.

System Requirements: Designed for Linux with NVIDIA GPUs using CUDA and PyTorch. CPU-only operation **in theory** is not feasible for the current templates due to computational demands. I ran it on Intel i5 processor.

Usage: Users can run experiments by setting up the environment, preparing data, and executing scripts through command line interfaces. It supports various models like **GPT-4o** and **Claude Sonnet 3.5**, **requiring** respective **API keys** and **pre-paid credits**.

Safety and Deployment: The project includes a **Docker image** for safer execution.

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Agentic AI – 2025/2026

Provider	AI Agent
Anthropic	Claude Code
Google	Gemini CLI
OpenAI	Codex
XAI	Grok CLI
Mistral	Mistral Vibe CLI

AI Workflow <= 2024

Copilot	Browser-based AI Chatbot
# Comment	Prompt
↓	↓
Code	Output
	↓
	CTRL-C CTRL-V
	↓
	Local file

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Agentic AI Workflow >= 2024

Actions
Literature Survey, Experiment Design, Execution, Evaluation, Write-up

Context/Planning
Queues e.g. FIFO, prompt history, log, memory

Tools
Semantic Scholar/ArXiv API/MCP Python/Bash Latex

Safety Net
Regex, grep – check GenAI/LLM text/string exist in source

Agent Agnostic
Well established workflows that work with any provider

Coder becomes Manager (Industry trend)
Directs agents to write, queue and execute prompts

Case Study – Clinical AI Hackathon

MOD/NHS Clinical pain point

Structure semi-structured medical notes

St George's Healthcare NHS
NHS Trust

Colorectal Multidisciplinary Meeting 07/03/2025(i)

Patient Details	Cancer Target Dates
Hospital Number: 9990001(d) NHS Number: 9990000001(c)	62 DAY TARGET: 20/05/2025

A	B	C	D	E	F	G	H
Demographics: DOB(a)	Demographics: Initials(b)	Demographics: MRN(c)	Demographics: NHS number(d)	Demographics: Gender(e)	Demographics: Previous cancer (y, n) if yes, where(f)	Demographics: State site of previous cancer(f)	Endoscopy: dat
26/05/1970	AO	9990001	9990000001	Male	No	N/A	Missing

Training and baseline solution provided

To be improved, with the use of AI Agents, by participating teams

Numbers

76 sign-ups (majority PG), 32 finalists, 9 teams

Outcomes

Generated two research placements at St George's NHS Foundation Trust

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Research Projects at St George's NHS Foundation Trust

Monitoring AI for Performance Drift: Developing an independent quality assurance framework to identify when radiology AI performance changes over time, enabling early detection of reliability issues in clinical practice.

Evaluating Subgroup Bias: Developing an independent assessment framework to identify whether radiology AI performs differently across patient groups, supporting fair, safe, and accountable clinical deployment.

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Clinical AI Hackathon – Lessons learnt

Healthcare is highly-regulated
Solutions are much easier to write than to deploy (certification)

Solutions inspired higher trust in Generative AI
Clinicians confident enough to agree on further deployment of Agentic AI in research

The sky is the limit
"I think entrepreneurs should be more ambitious about the problems they're going after. If the AI capability you need doesn't exist today, there's a reasonable chance it will in three to six months. So the limiting factor really shouldn't be what the technology can do right now."
Dario Amodei

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What happens next

ElectroAI Hackathon
City St George's, University of London 07-10.04.2026

British Computer Society-backed AI Levelling-up Hackathon
City St George's, University of London 30.06-01.07.2026

Learning at City St George's Conference 2026 – Snap Circuits Workshop
City St George's, University of London 30.06.2026

Slides, contacts
Please scan QR code or visit
https://github.com/dsikar/csg-ai-research-day-2026